



A game theoretical model of deforestation in human–environment relationships

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ABSTRACT

We studied a two-person game regarding deforestation in human–environment relationships. Each landowner manages a single land parcel where the state of land-use is forested, agricultural, or abandoned. The landowner has two strategies available: forest conservation and deforestation. The choice of deforestation provides a high return to the landowner, but it degrades the forest ecosystem services produced on a neighboring land parcel managed by a different landowner. Given spatial interactions between the two landowners, each landowner decides which strategy to choose by comparing the expected discounted utility of each strategy. Expected discounted utility is determined by taking into account the current and future utilities to be received, according to the state transition on the two land parcels. The state transition is described by a Markov chain that incorporates a landowner's choice about whether to deforest and the dynamics of agricultural abandonment and forest regeneration. By considering a stationary distribution of the Markov chain for land-use transitions, we derive explicit conditions for Nash equilibrium. We found that a slow regeneration of forests favors mutual cooperation (forest conservation). As the forest regenerates faster, mutual cooperation transforms to double Nash equilibria (mutual cooperation and mutual defection), and finally mutual defection (deforestation) leads to a unique Nash equilibrium. Two different types of social dilemma emerge in our deforestation game. The stag-hunt dilemma is most likely to occur under an unsustainable resource supply, where forest regenerates extremely slowly but agricultural abandonment happens quite rapidly. In contrast, the prisoner's dilemma is likely under a persistent or circulating supply of resources, where forest regenerates rapidly and agricultural abandonment occurs slowly or rapidly. These results show how humans and the environment mutually shape the dilemma structure in forest management, implying that solutions to dilemmas depend on environmental properties.

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1. Introduction

The transformation of forested lands by human activities is one of the great forces in global environmental change and biodiversity loss (Brooks et al., 2003; Curran et al., 2004; Dixon et al., 1994; Dobson et al., 1997; Fearnside, 2000). Extensive deforestation may increase the vulnerability and erode the resilience of ecophysical systems, which may result in far-reaching consequences for human well-being (Folke, 2006; Millennium Ecosystem Assessment, 2005; Turner et al., 2003; Vitousek et al., 1997).

Rapid population growth, increasing industrial demand, and poverty have been reported to be important drivers of deforestation

(Lambin et al., 2001; Laurance, 2007). However, the ultimate driver of deforestation is the human who chooses deforestation rather than forest conservation due to changes in socio-economic and ecological circumstances. Understanding the nature of people's decision-making about deforestation is crucial for an in-depth understanding of the causes of deforestation and for policy making to deter further deforestation.

Game theoretical models have long been used to analyze decision making in the management of common-pool resources. Common-pool resources are characterized by the difficulty of excluding actors from using them (excludability) and by the fact that their exploitation by one user reduces resource availability for others (subtractability) (Ostrom, 1990; Ostrom et al., 1999). For example, the prisoner's dilemma game with a large number of players lies at the heart of “the tragedy of the commons” proposed in Garrett Hardin's influential article (Hardin, 1968). This study

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has provided the metaphor for the realization that the strategic behavior of users of common-pool resources can cause inexorable tragedy and that users are unable to engage in sufficient collective action to extract themselves from drastic overuse and destruction, causing social dilemma. Hardin has argued that privatization and regulation by the state government can be solutions to the social dilemma.

However, there are many ecosystem services that are not easily parceled up. If the scale of provision for ecosystem services, such as pollinations of crops, purification of water, and stabilization of climate, were larger than the scale of decision making by actors who manage resources, scale mismatch problems would occur (Cumming et al., 2006; Satake et al., 2008). For example, land conversion from forest to agriculture provides benefit to local actors, but it may threaten water quality for people downstream by leaching fertilizers and pesticides (WRI, 2000). Forest harvests in upper river basins often result in serious soil erosion and floods downstream (Liu and Diamond, 2005). In scale mismatch problems, clear property right exists (generally excludable), but availability of ecosystem services are reduced by exploitation by others, generating spatial interactions among different management units with respect to their interests; thus, the landowner must anticipate the actions of his/her neighbor(s) to make appropriate decisions about his/her own resource use. Although this situation fits perfectly in game theory, the use of game theoretical models in scale mismatch problems in forest management is rare, unlike the case in common-pool resource problems.

In this study, we formulated and analyzed a two-player game about deforestation in which deforestation by one landowner produced a high return to that landowner, but degraded the forest ecosystem services produced at a neighboring land parcel managed by a different landowner. We are especially interested in developing a game theoretical model that couples choice behavior by landowners and ecological dynamics. We investigate situations where the deforestation game entails social dilemmas and what kinds of dilemmas they are. In this game, each land parcel is either in a forested, agricultural, or abandoned state. The state transition is described by a Markov chain that incorporates landowners' choices between forest conservation and deforestation as well as the dynamics of agricultural abandonment and of forest regeneration. Landowners make decisions about the choice between forest conservation and deforestation by taking into account the opponent's choice and future state changes in two adjacent land parcels.

By analyzing conditions for Nash equilibrium, we show how ecological factors (e.g., rate of forest regeneration) and socio-economic factors (e.g., income from crop production or forest conservation) influence the type and likelihood of social dilemmas. We present implications for management policies, considering different types of social dilemmas arising from our deforestation game.

2. The game theoretical model for deforestation

We assume two land parcels, each of which is individually managed by landowner i with $i \in \{1, 2\}$. Each landowner may be a household, a community, or a state government. Each landowner has two strategies, forest conservation and deforestation, defined as $r_i = 0$ and 1, respectively, and each decides which strategy to choose by comparing the expected discounted utility of the two strategies. Let $v(r_i|r_j)$ be the expected discounted utility of strategy r_i when the opponent chooses strategy r_j . The quantity $v(r_i|r_j)$ is determined by taking into account the current land-use state as well as the state change that may occur at the two land parcels in the future, as we define later.

In this section, we first explain the state transition dynamics of a single land parcel. Second, we define the snapshot utility given to each land-use state. Third, we define the expected discounted utility. We build on land-use models studied by Satake and Iwasa (2006), Satake et al. (2007), and Satake and Rudel (2007), and modify these models to fit into a game theoretical setting. In their models, choice behavior of landowners is described by logit dynamics (i.e., probabilistic decision), but in the model analyzed in the paper, landowners always choose the option that produces the largest utility.

2.1. State transition: forested (F), agricultural (A), and abandoned (E) land

Let $S_i(t)$ be the state variable of the land parcel i in year t . We consider three land-use states:

$$S_i(t) = \begin{cases} F & \text{if parcel } i \text{ is forested} \\ A & \text{if parcel } i \text{ is agricultural} \\ E & \text{if parcel } i \text{ is abandoned.} \end{cases} \quad (1)$$

The forested parcel is converted to an agricultural parcel with probability r_i (Fig. 1). We call r_i the “deforestation rate.” As defined above, $r_i = 0$ if the landowner i plays forest conservation; otherwise $r_i = 1$. For simplicity, we focus our analysis only on pure strategies. Agricultural land changes to abandoned land with probability η in a year (Fig. 1). We call η the “abandonment rate”. A small η represents the situation where agricultural operation continues to produce profits for many years. In contrast, a large η indicates that soil fertility and agricultural profits decline sharply with continued cultivation. Abandonment of agricultural land may result in the growth of secondary vegetation on an abandoned parcel. We assume that an abandoned parcel will finally revert back to forested land, with probability μ in a year (Fig. 1). We call μ the “forest recovery rate” per year. Natural regrowth of vegetation following agriculture or pasture abandonment has been reported in the Amazon (Moran et al., 1996), the island of Puerto Rico (Aide et al., 2000; Grau et al., 2003; Pascarella et al., 2000), and globally (Rudel et al., 2005).

2.2. Snapshot utilities

A snapshot utility is given to each land-use state. It is influenced by the land-use state of the opponent's parcel. Let $u(S_i|S_j)$ be the snapshot utility received by landowner i when his land parcel is in state S_i and the opponent's land parcel is in state S_j . We have a series of assumptions about $u(S_i|S_j)$, as follows (see Table 1 for summary).

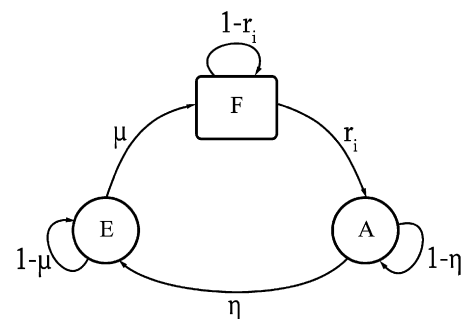


Fig. 1. A three-state cyclic Markov chain for land-use dynamics at a single land parcel. We have three states: forested (F), where the transition rates depend on the strategy adopted by the landowner; agricultural (A) and abandoned (E), where the transition rates depend on natural events.

Table 1
State-dependent snap-shot utilities.

	F	A	E
F	b, b	$b-\varepsilon, c$	$b-\varepsilon, 0$
A	$c, b-\varepsilon$	$c-\delta, c-\delta$	$c-\delta, 0$
E	$0, b-\varepsilon$	$0, c-\delta$	$0, 0$

The state F has a base utility of b . It is degraded by ε if the opponent's parcel is either in state A or E . State A has a base utility of c . It is degraded by δ if the opponent's parcel is either in state A or E . State E has zero utility.

- (1) Utility of healthy forested land: when the land-use states in both land parcels are forested, landowners receive utility of b , i.e., $u(F|F) = b$. The utility is caused by consumption of non-timber forest products (e.g., firewood, medical products, and wild fruits) or by a subsidy from the federal government.
- (2) Utility of degraded forested land: when the land-use state in the opponent's parcel is agricultural or abandoned, the utility of the forested parcel managed by the focal landowner is degraded by ε , i.e., $u(F|A) = u(F|E) = b - \varepsilon$, because parts of the forest ecosystem service are lost due to land conversion from forest to agriculture in the neighboring parcel. For example, a reduction in forest cover increases the rate of soil erosion and the leaching of soil nutrients (Hooke, 2006; Likens et al., 1970; Williams et al., 1997), degrading the growth and survival rates of trees in remaining forests. Note that $\varepsilon \geq 0$.
- (3) Utility of healthy agricultural land: When the land parcel i is in an agricultural state and when the opponent's land parcel is forested, the landowner i receives utility c from agricultural operations, i.e., $u(A|F) = c$, where c is the utility of agriculture, which is the total revenue received (e.g., monetary benefits from crop sales) minus the costs incurred (e.g., cultivating, harvesting, and transportation costs).
- (4) Utility of degraded agricultural land: When the land parcel i is in an agricultural state and the opponent's land parcel is agricultural or is abandoned, the utility of the forested parcel managed by the focal landowner is degraded by δ , i.e., $u(A|A) = u(A|E) = c - \delta$. This assumption comes from the fact that the loss of forest has a notable effect on the degradation of crop pollination in neighboring forests, through reduced pollinator diversity and abundance (Aizen and Feinsinger, 1994; Klein et al., 2007). Note that $\delta \geq 0$.
- (5) The utility of an abandoned parcel is zero, lower than that of healthy forested and healthy agricultural lands, because bare land does not produce any ecosystem services.

2.3. Expected discounted utilities

2.3.1. Expected discounted utilities as a function of initial land-use state

We first define the expected discounted utility as a function of the initial land-use state. Let $V(S_k, r_i|S_l, r_j)$ be the conditional expected discounted utility of the strategy r_i when the opponent plays strategy r_j and when the land parcels are in states S_k for the focal parcel i and S_l for the parcel j . We define $V(S_k, r_i|S_l, r_j)$ as follows:

$$V(S_k, r_i|S_l, r_j) = \sum_{n=0}^{\infty} \omega^n U_n^i(S_k, r_i|S_l, r_j), \quad (2)$$

where $U_n^i(S_k, r_i|S_l, r_j)$ is the expected utility to be received in year n in the future if the focal parcel i is in state S_k and the opponent's parcel is in state S_l at the present time. The land-use state may change in the future, and thus $U_n^i(S_k, r_i|S_l, r_j)$ includes all possible contributions from different combinations of land-use states on two land parcels, as explained later. The quantity ω is the discount

factor ($0 \leq \omega < 1$). When ω is close to 1, the landowner identifies the future utility of the parcel as being as important as the current utility. In contrast, if ω is close to 0, the landowner attaches greater importance to the current utility. The landowner i who plays a strategy r_i employs the expectation for the future landscape change at a parcel i represented by the Markov transition matrix $\mathbf{P}(r_i)$, as follows (Satake and Iwasa, 2006):

$$\mathbf{P}(r_i) = \begin{pmatrix} 1-r_i & r_i & 0 \\ 0 & 1-\eta & \eta \\ \mu & 0 & 1-\mu \end{pmatrix}. \quad (3)$$

As we defined in Table 1, the snapshot utility is affected by the land-use state of the opponent's land parcel as well as that of the focal land parcel. Thus, we need to take into account the state transition of the opponent's land parcel in addition to that in the focal land parcel. We represent this interrelated state change at two parcels by introducing a transition matrix $\mathbf{A}(r_i, r_j)$, where $\mathbf{A}(r_i, r_j)$ is now a 9×9 matrix given by the Kronecker product of the transition matrix at parcels i and j , as follows (Satake et al., 2007):

$$\mathbf{A}(r_i, r_j) = \mathbf{P}(r_i) \otimes \mathbf{P}(r_j), \quad (4)$$

where $\mathbf{P}(r_i)$ is defined in Eq. (3) and $\mathbf{P}(r_j)$ is defined correspondingly.

Let $\mathbf{u} = (u(F|F), u(F|A), u(F|E), \dots, u(E|E))^T$ be the column vector of snapshot utilities for the nine possible combinations of land-use states in the two land parcels. Let $\mathbf{U}_n^i(r_i, r_j) = (U_n^i(F, r_i|F, r_j), U_n^i(F, r_i|A, r_j), \dots, U_n^i(E, r_i|E, r_j))^T$ be a column vector composed of the expected utilities to be received by landowner i after n time steps in the future for the nine possible combinations of initial land-use states in the two land parcels (in the following, we remove superscript i and the dependency on the strategies). The vector $\mathbf{U}_n$ changes according to $\mathbf{U}_{n+1} = \mathbf{A}\mathbf{U}_n$. Because the utility to be received at the present time, $\mathbf{U}_0$, is simply given by the snapshot utility vector $\mathbf{u}$, we have the following relationship: $\mathbf{U}_n = \mathbf{A}^n \mathbf{u}$. Using this relationship, the expected discounted utilities for each combination of initial land-use states, denoted by $\mathbf{V}(r_i, r_j) = (V(F, r_i|F, r_j), V(F, r_i|A, r_j), V(F, r_i|E, r_j), \dots, V(E, r_i|E, r_j))^T$, is given as:

$$\mathbf{V}(r_i, r_j) = \sum_{n=0}^{\infty} [\omega \mathbf{A}(r_i, r_j)]^n \mathbf{u}. \quad (5)$$

In principle, based on the conditional expected discounted utilities defined in Eq. (5), we can derive the Nash equilibrium of the deforestation game. However, we found that conditions for Nash equilibrium derived from each potential pair of initial land-use states were too complicated to provide useful insight about forest management (see Appendix A). Thus, we here focus our analysis on when the discount factor, ω , is close to 1.

2.3.2. When the discount factor is close to 1

When the discount factor is close to 1, the probability that the land-use state at a single parcel is in state S_i is approximated by the stationary distribution of the Markov chain. Let $\mathbf{q}_i^*(r_i) = (q_{F,i}^*(r_i), q_{A,i}^*(r_i), q_{E,i}^*(r_i))^T$ be the stationary distribution of the Markov chain for the land-use transitions at a land parcel i . Given a strategy r_i for a landowner i , $\mathbf{q}_i^*(r_i)$ is equivalent to the eigenvector of the transition matrix $\mathbf{P}(r_i)$, which corresponds to an eigenvalue of 1:

$$q_{F,i}^*(r_i) = \frac{\eta\mu}{r_i\eta + r_i\mu + \eta\mu}, \quad q_{A,i}^*(r_i) = \frac{r_i\mu}{r_i\eta + r_i\mu + \eta\mu} \quad \text{and} \quad q_{E,i}^*(r_i) = \frac{r_i\eta}{r_i\eta + r_i\mu + \eta\mu}. \quad (6)$$

Using Eq. (6), for instance, the probability that both parcels are forested under strategy profile $\{r_1, r_2\}$ is given by $q_{F,1}^*(r_1)q_{F,2}^*(r_2)$. Thereafter, the expected discounted utility under ω close to 1 is given by the expected value of the conditional expected discounted

utility introduced in Eq. (5):

$$v(r_1|r_2) = \sum_{k=1}^3 \sum_{l=1}^3 q_{s_k,1}^*(r_1) q_{s_l,2}^*(r_2) V(s_k, r_1 | s_l, r_2) \quad \text{with} \\ s_1 = F, \quad s_2 = A, \quad s_3 = E. \quad (7)$$

In the following, we determine the Nash equilibrium of our deforestation game using Eq. (7).

3. The Nash equilibrium

3.1. Definition

Because the snapshot utilities summarized in Table 1 are symmetric and the state transition is homogeneous between two land parcels, we obtain:

$$\begin{aligned} v(r_1 = 0|r_2 = 0) &= v(r_2 = 0|r_1 = 0) = v(0|0) \\ v(r_1 = 1|r_2 = 0) &= v(r_2 = 1|r_1 = 0) = v(1|0) \\ v(r_1 = 0|r_2 = 1) &= v(r_2 = 0|r_1 = 1) = v(0|1) \\ v(r_1 = 1|r_2 = 1) &= v(r_2 = 1|r_1 = 1) = v(1|1). \end{aligned} \quad (8)$$

Consider the strategy profile $\sigma_{r_1 r_2} = \{r_1, r_2\}$. The strategy profile $\sigma_{r_1 r_2}$ is the Nash equilibrium if and only if

$$v(r_1|r_2) \geq v(1-r_1|r_2) \quad \text{and} \quad v(r_2|r_1) \geq v(1-r_2|r_1). \quad (9)$$

We summarize conditions for Nash equilibrium for the deforestation game:

(1) $\hat{\sigma}_{00}$: The strategy profile, (forest conservation, forest conservation), is the Nash equilibrium when

$$v(0|0) \geq v(1|0). \quad (10)$$

(2) $\hat{\sigma}_{10} \wedge \hat{\sigma}_{01}$: The strategy profiles, (deforestation, forest conservation) and (forest conservation, deforestation), are the Nash equilibrium when

$$v(1|0) \geq v(0|0) \wedge v(0|1) \geq v(1|1). \quad (11)$$

(3) $\hat{\sigma}_{11}$: The strategy profile, (deforestation, deforestation), is the Nash equilibrium when

$$v(1|1) \geq v(0|1). \quad (12)$$

Based on the inequalities given above, we derive the explicit conditions for Nash equilibrium and discuss how ecological parameters, such as agricultural abandonment and forest recovery rates, affect the equilibrium.

3.2. Conditions and classifications of the Nash equilibrium

From Eqs. (6) and (7), we have

$$v(0|0) = \frac{b}{(1-\omega)}, \quad (13a)$$

$$v(0|1) = \frac{1}{(1-\omega)} \left[b - \frac{\varepsilon(\eta + \mu)}{(\eta + \mu + \eta\mu)} \right], \quad (13b)$$

$$v(1|0) = \frac{1}{(1-\omega)(\eta + \mu + \eta\mu)} [\eta\mu b + \mu c], \quad (13c)$$

$$\begin{aligned} v(1|1) &= \frac{1}{(1-\omega)(\eta + \mu + \eta\mu)} \\ &\times \left[\eta\mu b + \mu c - \frac{(\eta + \mu)(\mu\delta + \eta\mu\varepsilon)}{(\eta + \mu + \eta\mu)} \right]. \end{aligned} \quad (13d)$$

The above equations illustrate how the transition dynamics of land-use state (the rates of agricultural abandonment, η , and forest regeneration, μ) as well as snapshot utilities (b , c , ε , δ)

influence the payoff structure of the deforestation game. The structural changes in payoffs alter the nature of the dilemma, even determining whether a social dilemma exists at all. Applying Eqs. (13) to the conditions for the Nash equilibrium given in Eqs. (10)–(12) leads to the following:

(1) $\hat{\sigma}_{00}$ is the Nash equilibrium when

$$b \geq \frac{\mu}{\eta + \mu} c. \quad (14)$$

The above condition means that the strategy profile, (forest conservation, forest conservation), is more likely to be the Nash equilibrium as forests regenerate more slowly and as landowners abandon their agricultural land more rapidly.

(2) $\hat{\sigma}_{10}$ and $\hat{\sigma}_{01}$ are the Nash equilibria when:

$$b \leq \frac{\mu}{\eta + \mu} c \wedge b \geq \frac{\mu}{\eta + \mu} c + \frac{\varepsilon(\eta + \mu) - \mu\delta}{(\eta + \mu + \eta\mu)}. \quad (15)$$

(3) $\hat{\sigma}_{11}$ is the Nash equilibrium when:

$$b \leq \frac{\mu}{\eta + \mu} c + \frac{\varepsilon(\eta + \mu) - \mu\delta}{(\eta + \mu + \eta\mu)}. \quad (16)$$

The partial derivative of the right-hand side in Eq. (16) by μ and η shows that the strategy profile, (deforestation, deforestation), is more likely to be the Nash equilibrium as the forest recovery rate increases and as the rate of agricultural abandonment decreases. Note that the discount factor (ω) disappeared because in Eqs. (13) all terms are affected equally by $(1-\omega)$. In the following, we use the terms defined below for simplicity:

$$\phi \equiv \frac{\mu}{\eta + \mu} c, \quad \psi \equiv \frac{\varepsilon(\eta + \mu) - \mu\delta}{\eta + \mu + \eta\mu}. \quad (17)$$

Based on the three inequalities in Eqs. (14)–(16), we classify the equilibrium depending on the sign of $\varepsilon(\eta + \mu) - \mu\delta$ that appears on the right-hand side in Eqs. (15) and (16).

Case I: Degradation of forest ecosystem services is severe: $\varepsilon \geq \mu\delta/(\mu + \eta)$. Case I corresponds to the situation in which deforestation leads to a more severe degradation of forest ecosystem services (ε) compared with the extent of the degradation in agricultural activity (δ). The magnitude of δ is weighted by the factor $\mu/(\mu + \eta)$.

We illustrate how the equilibrium changes according to the utility of the forested parcel (b ; Fig. 2). Three different cases exist, depending on the relative magnitudes of b , ϕ , and ψ (Fig. 2a).

(I-1) $\hat{\sigma}_{00}$, (forest conservation, forest conservation), is the unique Nash equilibrium when $b > \phi + \psi$.

(I-2) $\hat{\sigma}_{11}$, (deforestation, deforestation), is the unique Nash equilibrium when $b < \phi$.

(I-3) There are two equilibria, $\hat{\sigma}_{00}$ and $\hat{\sigma}_{11}$, when $\phi \leq b \leq \phi + \psi$.

Case II: Degradation of forest ecosystem services is moderate: $\varepsilon < \mu\delta/(\mu + \eta)$. Case II corresponds to the situation in which degradation of forest ecosystem services due to deforestation (ε) is moderate compared with the extent of degradation of agricultural utility (δ). In this case, we obtained slightly different results from case I (Fig. 2b):

(II-1) $\hat{\sigma}_{00}$ is the unique Nash equilibrium when $b > \phi$.

(II-2) $\hat{\sigma}_{11}$ is the unique Nash equilibrium when $b < \phi + \psi$.

(II-3) There are two equilibria, $\hat{\sigma}_{01}$ and $\hat{\sigma}_{10}$, when $\phi + \psi \leq b \leq \phi$.

Note that ψ is negative in the case of II-2. In the case of II-3, a landowner always tries to choose the strategy opposite to that of his/her opponent because the same choice (deforestation, deforestation) results in significant deterioration of agricultural utility.

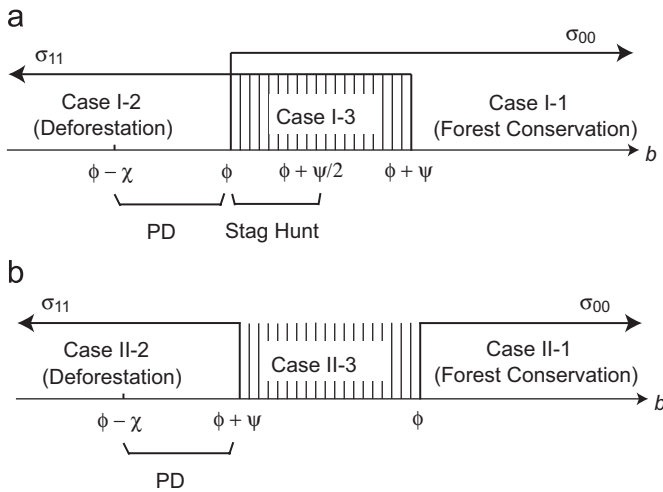


Fig. 2. (a) If $\varepsilon \geq \mu\delta/(\mu + \eta)$, there are three situations for the Nash equilibrium. If $b > \phi + \psi$, (forest conservation, forest conservation) is the unique Nash equilibrium. If $b < \phi$, (deforestation, deforestation) is the unique Nash equilibrium. If $\phi \leq b \leq \phi + \psi$, both strategy profiles, (forest conservation, forest conservation) and (deforestation, deforestation), are the Nash equilibria. (b) If $\varepsilon < \mu\delta/(\mu + \eta)$, there are three situations for the Nash equilibrium. If $b > \phi$, (forest conservation, forest conservation) is the unique Nash equilibrium. If $b < \phi + \psi$, (deforestation, deforestation) is the unique Nash equilibrium. If $\phi + \psi \leq b \leq \phi$, both strategy profiles, (forest conservation, deforestation) and deforestation, forest conservation, are the Nash equilibria.

In our deforestation game, not all situations entail dilemmas. For such a situation to be a dilemma, it must be demonstrated that suboptimal outcomes can occur if landowners attempt to maximize their personal interests. In the following, we investigate situations where the deforestation game entails social dilemmas and what kinds of dilemmas they are. For that purpose, we first study equilibrium selection.

4. Equilibrium selection

In case (I-3) in the previous section, we found two Nash equilibria, $\hat{\sigma}_{00}$ and $\hat{\sigma}_{11}$. Nash equilibrium is a major solution concept in game theory (Nash, 1950; Fudenberg and Tirole, 1991; Binmore, 2007; Gintis, 2000). When there exists the only one equilibrium, we expect that it should be realized as the outcome of conflicts. However, when there are multiple equilibria we wonder which equilibrium is actually realized. This problem is called “equilibrium-selection” and has been intensively studied by game theorists for decades (Gintis, 2000; Binmore, 2007).

In case (I-3), the equilibrium $\hat{\sigma}_{00}$ corresponds to the situation in which both landowners choose forest conservation strategy, while both landowners choose deforestation strategy at the equilibrium $\hat{\sigma}_{11}$. What happens if two landowners try to maximize the sum of their payoffs? The sum of payoff is $2v(0|0)$ for the former equilibrium, and $2v(1|1)$ for the latter. Therefore two landowners would compare $v(0|0)$ and $v(1|1)$, and would choose whichever gives the largest. The equilibrium that yields the larger collective payoff than the other is called payoff-dominant equilibrium (Harsanyi and Selten, 1988). In case (I-3), $v(0|0) > v(1|1)$ always holds (in fact, $v(0|0) > v(1|1)$ holds when $b > \phi - \chi$, where $\chi = \mu(\eta\varepsilon + \delta)/(\eta + \mu + \eta\mu)$). Therefore, the forest-conservation equilibrium is payoff-dominant, and two landowners would choose forest conservation strategy if they try to maximize the collective payoff.

However, since each landowner can act selfishly to maximize his/her individual payoff, we cannot expect that they should choose the payoff-dominant equilibrium. Then we come back to our original question: which equilibrium is chosen?

We can give one answer to this question by comparing the stability of two equilibria. For that purpose we extend our original model that involves only two landowners to the one with (infinitely) many landowners. Each landowner meets another randomly to play the original two-person deforestation game. This assumption can be justified if landowners manage multiple land parcels under the same policy and if these land parcels are randomly distributed in landscape. Let x_0 or x_1 be the fraction of landowners who adopt forest conservation or deforestation strategy, respectively. We have $x_0 + x_1 = 1$. The expected utility of each strategy is given by

$$E_0(x_0, x_1) = v(0|0)x_0 + v(0|1)x_1, \quad (18)$$

$$E_1(x_0, x_1) = v(1|0)x_0 + v(1|1)x_1. \quad (19)$$

Since the forest conservation strategy is a Nash equilibrium, if *all* landowners adopt the forest conservation strategy, then adopting the forest conservation is better than the deforestation strategy: namely,

$$E_0(1, 0) > E_1(1, 0). \quad (20)$$

By relaxing “*all* landowners” in the previous sentence, we ask until when the forest conservation strategy remains better than the deforestation strategy. Imagine that *the fraction of p of landowners* adopts the forest conservation strategy but the others adopt the deforestation strategy. In this situation, the forest conservation strategy is the better if

$$E_0(p, 1 - p) > E_1(p, 1 - p). \quad (21)$$

Solving Eq. (21) gives

$$p > p_0 \equiv \frac{v(1|1) - v(0|1)}{\{v(0|0) - v(1|0)\} + \{v(1|1) - v(0|1)\}}. \quad (22)$$

The threshold p_0 in Eq. (22) represents the minimum fraction of landowners who adopts the forest conservation strategy in order for the forest conservation strategy to be better than the deforestation strategy. In other words, the size of p_0 reflects the stability of the forest conservation equilibrium: the smaller p_0 is, the more stable the equilibrium is.

Similarly, we study the stability of the deforestation-equilibrium. Since it is the Nash equilibrium, we have

$$E_1(0, 1) > E_0(0, 1). \quad (23)$$

When *the fraction of p of landowners* adopts the deforestation strategy, solving $E_1(1 - p, p) > E_0(1 - p, p)$ yields

$$p > p_1 \equiv \frac{v(0|0) - v(1|0)}{\{v(0|0) - v(1|0)\} + \{v(1|1) - v(0|1)\}}. \quad (24)$$

From these results, we can compare the stability of the two equilibria. The forest conservation equilibrium is more stable than the deforestation one when $p_0 < p_1$, which is simplified to

$$v(0|0) + v(0|1) > v(1|0) + v(1|1). \quad (25)$$

The inequality (25) is well-known as the condition for *risk-dominance* (Harsanyi and Selten, 1988; Gintis, 2000; Binmore, 2007). Strategy 0 (forest conservation) is called *risk-dominant* over strategy 1 (deforestation) when Eq. (25) holds. Intuitively the *risk-dominant* strategy is the one that leads to the larger payoff when the opponent chooses the two strategies equally likely. Notably, the condition for *risk-dominance*, Eq. (25), does not agree with the one for *payoff-dominance*, $v(0|0) > v(1|1)$. It suggests that the more efficient outcome for the group of landowners does not necessarily mean the more stable outcome. In fact, Kandori et al. (1993) and Young (1993) have shown that under the population dynamics in which players try to maximize one's individual payoff, *risk-dominant* equilibrium is most likely to be chosen.

By applying Eq. (25) to our deforestation game, we obtain the following result. Remember that case (I-3) occurs when $\phi \leq b \leq \phi + \psi$.

- (i) When $\phi \leq b < \phi + \psi/2$, the deforestation strategy is *risk-dominant*. Therefore the deforestation equilibrium is more stable and more likely to be chosen, although the forest conservation strategy is *payoff-dominant* and therefore it is socially more profitable. This is a dilemmatic-situation.
- (ii) When $\phi + \psi/2 < b \leq \phi + \psi$, the forest conservation strategy is *risk-dominant*. Therefore the forest conservation equilibrium is more stable and more likely to be chosen. As it is also *payoff-dominant*, no social dilemma arises.

5. Resource dynamics influence the likelihood of social dilemma

When deforestation enters into an environmental problem, it should be characterized by a social dilemma in which private interests (e.g., profit from deforestation) are at odds with the social optimum (e.g., forest conservation). To understand when a social dilemma emerges in our deforestation game, we compared the deforestation game with two well-known social dilemma games: the stag hunt game, where both mutual cooperation and mutual defection are the Nash equilibria (Skyrms, 2004); and the prisoner's dilemma game, where mutual defection of players is the unique Nash equilibrium (Axelrod and Hamilton, 1981).

The stag hunt game is characterized by the following inequalities: $v(0|0) > v(1|0) \geq v(1|1) > v(0|1)$. From Eq. (13), it is easily shown that the deforestation game has the same properties as the stag hunt game when $\phi < b < \phi + \psi$, which fits within the parameter region of case I-3. In case I-3, a social dilemma emerges when $\phi \leq b < \phi + \psi/2$ because the *payoff-dominant* strategy is not consistent with the *risk-dominant* strategy, as shown in the previous section. The likelihood of this dilemma is independent of snapshot utilities of forested (b) and agricultural land (c), but rather depends on the quantity of ψ , defined in Eq. (17). ψ is a decreasing function of the forest recovery rate (μ), and it attains the maximum of ε when $\mu = 0$, regardless of the magnitude of η . The rate of agricultural abandonment (η) influences the likelihood of the dilemma in a more complicated way. The partial derivative of ψ by η shows that $\partial\psi/\partial\eta > 0$ when $\varepsilon < (1 + 1/\mu)\delta$, while $\partial\psi/\partial\eta < 0$ when $\varepsilon > (1 + 1/\mu)\delta$. When $\mu \ll 1$, $\varepsilon < (1 + 1/\mu)\delta$ is always satisfied. Thus, the society is most likely to face the stag hunt dilemma when forest resources recover slowly and agricultural abandonment happens rapidly. In other words, the dilemma arises when the duration of the profitable state (i.e., the agricultural state) is the shortest, and that of the unprofitable state (i.e., the abandoned state) is the longest. We call this situation “an unsustainable supply of resources.”

In contrast, the prisoner's dilemma game is characterized by $v(1|0) > v(0|0) > v(1|1) > v(0|1)$. The strategy profile (deforestation, deforestation) is the unique Nash equilibrium. The deforestation strategy is a dominant strategy; thus, players will play for the Nash equilibrium even though both playing forest conservation strategy yields a larger collective payoff, i.e. $v(0|0) > v(1|1)$. Substituting the values of the expected discounted utilities defined in Eq. (13), we find that the deforestation game has the same properties as the prisoner's dilemma game if

$$\begin{cases} \phi - \chi < b < \phi & \text{if } \varepsilon \geq \mu\delta/(\eta + \mu), \\ \phi - \chi < b < \phi + \psi & \text{otherwise,} \end{cases} \quad (26)$$

where χ is defined as $\mu(\eta\varepsilon + \delta)/(\eta + \mu + \eta\mu)$. The likelihood of the prisoner's dilemma situation depends on the magnitude of the

quantity χ in the situation of case I (i.e., $\varepsilon \geq \mu\delta/(\eta + \mu)$; see Fig. 2a). χ is an increasing function of μ . The dependency of the agricultural abandonment rate is completely the opposite of the case in the stag hunt game. The partial derivative of χ by η shows that $\partial\chi/\partial\eta > 0$ when $\varepsilon > (1 + 1/\mu)\delta$, while $\partial\chi/\partial\eta < 0$ when $\varepsilon < (1 + 1/\mu)\delta$. When $\mu = 1$, χ attains the maximum of δ when $\eta = 0$ and $\delta < \varepsilon < 2\delta$, while it attains the maximum of $(\varepsilon + \delta)/3$ when $\eta = 1$ and $\varepsilon > 2\delta$. Thus, the society is most apt to face the prisoner's dilemma in the two following situations: (1) a persistent supply of resources; when $\delta < \varepsilon < 2\delta$, the society is most apt to face the prisoner's dilemma when forest resources recover rapidly and agricultural abandonment happens very slowly. (2) A circulating supply of resources; when $\varepsilon > 2\delta$, the society is most apt to face the prisoner's dilemma when both forest regeneration and agricultural abandonment happen rapidly. In both cases, if the rate of forest regeneration approaches zero, the prisoner's dilemma situation is resolved.

In the situation of case II (i.e., $\varepsilon < \mu\delta/(\eta + \mu)$), the likelihood of the prisoner's dilemma depends on the magnitude of $\psi + \chi$, which is simplified to ε . This means that the prisoner's dilemma situation is more likely to emerge as the extent of degradation of the forested parcel (ε) increases, regardless of the magnitude of η and μ .

6. Discussion

A major challenge of modern society is to find solutions to social dilemmas involving the use of limited forest resources. The purpose of this study was to develop and analyze a game theoretical model of deforestation to advance understanding of landowner's behavior in regard to deforestation in human–environment relationships. Our model emphasizes the interplay between human and environmental reactions, allowing us to show explicitly how environmental factors, such as the rate of forest regeneration, influence the type and likelihood of social dilemmas.

We derived the explicit conditions for Nash equilibrium of our two-player deforestation game and classified three primary outcomes: (1) forest conservation is the unique Nash equilibrium, (2) deforestation is the unique Nash equilibrium, and (3) both forest conservation and deforestation are Nash equilibria. Which outcome emerges depends critically on the rate of forest regeneration and that of agricultural abandonment, as well as on the snapshot payoffs of forest conservation and agriculture. When the forest recovers slowly and when snapshot payoffs of forest conservation are large, both landowners choose a forest conservation strategy; this choice meets the social optimum, implying there is no dilemma. Even with the slow regeneration of forests, when deforestation by one landowner causes severe degradation of a forest managed by the opponent (i.e., large ε), the choice of forest conservation strategy becomes frequency-dependent. In this situation, landowners choose a forest conservation strategy if others also choose forest conservation, otherwise they switch their strategies from forest conservation to deforestation. This situation causes a social dilemma when the *payoff-dominant* (i.e., forest conservation) is not chosen by an individual landowner, leading to a stag hunt-type dilemma. When forest regenerates rapidly, both landowners choose a deforestation strategy, resulting in another type of social dilemma akin to the prisoner's dilemma (Fig. 2). Thus, an increase in the forest regeneration rate transforms a stag hunt game to a prisoner's dilemma game.

How to resolve the social dilemma is the most important question we face. Our analysis of the deforestation game implies that policy intervention, such as subsidies for a forested parcel (an increase of b) and taxes for agricultural crops (a decrease of c)

would encourage a transition from deforestation to forest conservation behavior, extracting participants from the trap of a suboptimal situation. The likelihood of a social dilemma, however, is not influenced by a change in socio-economic factors, b and c . It is, rather, determined by environmental factors: the rates of forest regeneration and agricultural abandonment (μ and η), and the extent of degradation of forested and agricultural parcels due to deforestation (ε and δ). The likelihood of a stag hunt dilemma, measured by the parameter range where a stag hunt dilemma emerges (Fig. 2), attains the maximum under an unsustainable resource supply situation, where forest regenerates extremely slowly but agricultural abandonment happens quite rapidly. In contrast, the likelihood of a prisoner's dilemma is the greatest under a persistent or circulating supply of resources, where forest regenerates rapidly and agricultural abandonment occurs either slowly or rapidly.

These results may have important implications for forest management. In temperate forests where forests regenerate slowly, bottom-up regulations that enhance mutual trust between landowners, leading landowners to believe that others will choose a forest conservation strategy, may be effective in transforming deforestation into a forest conservation regime. On the other hand, in tropical forests where regeneration is rapid, top-down regulations, such as policy intervention to alter the economic value of forest conservation and deforestation, may be more useful to deter deforestation.

In this report, we show how humans and environment mutually shape the dilemma structure in forest management. Studies of coupled social and ecological systems are increasingly demanded in environmental sciences (Liu et al., 2007). We believe that our game theoretical theory model of deforestation in human environmental relationships provides an important contribution to this new field.

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Appendix A

We derive the conditions for Nash equilibrium when initial land-use states of two land parcels are both forested.

From Eq. (5) in the text, we have;

$$V(F, 0|F, 0) = \frac{b}{(1-\omega)}, \quad (\text{A.1})$$

$$V(F, 1|F, 0) = \frac{(1-\omega+\mu\omega)}{(1-\omega)(1-\omega+\eta\omega+\mu\omega+\eta\mu\omega^2)} \times [(1-\omega+\eta\omega)b+\omega c], \quad (\text{A.2})$$

$$V(F, 0|F, 1) = \frac{1}{(1-\omega)} \left[b - \frac{\omega(1-\omega+\eta\omega+\mu\omega)}{(1-\omega+\eta\omega+\mu\omega+\eta\mu\omega^2)} \varepsilon \right], \quad (\text{A.3})$$

$$V(F, 1|F, 1) = \frac{(1-\omega+\mu\omega)(\omega c + (1-\omega+\eta\omega))}{(1-\omega)(1-\omega+\eta\omega+\mu\omega+\eta\mu\omega^2)} + f(\varepsilon, \delta, \eta, \mu, \omega). \quad (\text{A.4})$$

Because $V(F, 1|F, 1)$ is too complicated, we just describe it using the function $f(\varepsilon, \delta, \eta, \mu, \omega)$. The strategy profile $\hat{\sigma}_{00}^{FF}$, (forest conservation, forest conservation), is the Nash equilibrium when:

$$V(F, 0|F, 0) \geq V(F, 1|F, 0). \quad (\text{A.5})$$

The strategy profile $\hat{\sigma}_{10}^{FF}$ or $\hat{\sigma}_{01}^{FF}$, (deforestation, forest conservation) or (forest conservation, deforestation), is the Nash equilibrium when:

$$V(F, 1|F, 0) \geq V(F, 0|F, 0) \wedge V(F, 0|F, 1) \geq V(F, 1|F, 1). \quad (\text{A.6})$$

The strategy profile, (deforestation, deforestation), is the Nash equilibrium when:

$$V(F, 1|F, 1) \geq V(F, 0|F, 1). \quad (\text{A.7})$$

Using Eqs. (A.1)–(A.4), we obtain the followings:

(1) $\hat{\sigma}_{00}^{FF}$ is the Nash equilibrium when:

$$b \geq \frac{(1-\omega+\mu\omega)}{(1-\omega+\eta\omega+\mu\omega)} c. \quad (\text{A.8})$$

(2) $\hat{\sigma}_{10}^{FF}$ and $\hat{\sigma}_{01}^{FF}$ are the Nash equilibria when:

$$b \leq \frac{(1-\omega+\mu\omega)}{(1-\omega+\eta\omega+\mu\omega)} c \wedge b \geq \frac{(1-\omega+\mu\omega)}{(1-\omega+\eta\omega+\mu\omega)} c + \frac{B^T \Theta \varepsilon - A^T \Theta \delta}{h}, \quad (\text{A.9})$$

where

$$A = \begin{pmatrix} -1 & \eta^2 - (2-\mu)\eta - \mu(3-\mu) + 2 \\ -(1-\mu+\mu^2)\eta^2 + (2-3\mu+3\mu^2-\mu^3)\eta - (1-\mu)^3 & [\eta - (1-\mu)^2]\eta\mu^2 \\ (1-\mu)(1-\eta-\mu)\eta^2\mu^2 & \end{pmatrix},$$

$$B = \begin{pmatrix} -1 & \eta^2 - (3-\mu)\eta - \mu(3-\mu) + 2 \\ -(1-\mu)^3\eta^3 + (2(2-\mu)\mu-3)\eta^2 + [3-\mu(5-(4-\mu)\mu)]\eta + 1 - \mu & -[\eta^3 - (2-\mu)\eta^2 + (1-3\mu+\mu^2)\eta + (1-2\mu+\mu^2)\mu]\eta\mu \\ (1-\eta-\mu)^2\eta^2\mu^2 & \end{pmatrix},$$

$$\Theta = \begin{pmatrix} 1 \\ \omega \\ \omega^2 \\ \omega^3 \\ \omega^4 \end{pmatrix},$$

$$h = -(1-\eta\mu\omega)[1-(1-\eta-\mu)\omega][1+\omega\{-(1-\mu)^2+2\eta-\eta^2(1-\mu^2\omega)\}].$$

(3) $\hat{\sigma}_{00}^{FF}$ is the Nash equilibrium when:

$$b \leq \frac{(1-\omega+\mu\omega)}{(1-\omega+\eta\omega+\mu\omega)} c + \frac{B^T \Theta \varepsilon - A^T \Theta \delta}{h}. \quad (\text{A.10})$$

When the discount factor ω approaches 1, the conditions in Eqs. (A.8)–(A.10) becomes the same with Eqs. (14)–(16) in the text.

We can apply the same argument for other pairs of initial states.

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